Comparative Analysis of Deep Neural Networks for Lung Nodule Segmentation: Performance and Insights

Arman Zafaranchi (669445) and Gurban Aliyev (664731)

University of Pisa, Pisa, Italy
{arman.zafaranchi, gurban.aliyev}@phd.unipi.it

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1 Introduction

Lung cancer is one of the most prevalent and deadly forms of cancer worldwide, accounting for a significant portion of cancer-related mortality. Early detection of lung cancer drastically improves the chances of successful treatment and patient survival. Computed tomography (CT) scans are a primary imaging modality used for the detection and diagnosis of lung abnormalities, including lung nodules, which can be indicative of malignancy.

Lung segmentation and nodule segmentation are two critical tasks in the automated analysis of lung CT scans. Lung segmentation involves identifying and isolating the lung regions from the surrounding anatomy and structures within the CT images. This step is essential as it defines the region of interest (ROI) where lung nodules are likely to be found, facilitating more focused and accurate nodule detection and analysis. Lung nodule segmentation specifically focuses on accurately identifying and delineating the nodules within the segmented lung regions. Accurate nodule segmentation is vital for assessing the size, shape, and growth of nodules, which are crucial parameters for diagnosing and staging lung cancer, planning treatment, and monitoring disease progression. Manual segmentation of lung regions and nodules by radiologists is labor-intensive and subject to inter-observer variability. Automated segmentation using deep learning techniques promises to enhance the efficiency, consistency, and accuracy of these tasks, providing robust support in clinical settings.

The main goal of this report is to apply various deep neural networks with different architectures to Lung CT scans for the segmentation of nodules. Also, the report aims to evaluate and compare the performance of three different neural network models for this task. Through this analysis, we seek to determine which model offers the highest accuracy and robustness in segmenting lung nodules from CT images. To achieve this, we first need to identify the region of interest (ROI), specifically the lung lobes within the CT images. By segmenting the lungs from the rest of the image, the subsequent and primary networks can concentrate on the nodules, ensuring that irrelevant sections do not affect the neural networks' performance.

In the initial step, we employed a pre-trained U-Net architecture that has previously been used for detecting lung lesions in COVID-19 cases. The segmented lungs obtained from this process were then applied to the raw CT scans. The resulting images were subsequently fed into different deep neural networks (DNNs) separately to assess their performance on the same dataset. We have selected three different architectures with similar bases, so we can observe the additional parts' effect on the basic performance of the DNN.

2 Data

For our study, we used the LUNA16 (Lung Nodule Analysis 2016) dataset[1] to train and evaluate different deep-learning models for lung nodule segmentation. The LUNA16 derives from the publicly available LIDC-IDRI (Lung Image Database Consortium and Image Database Resource Initiative) database[2] and includes a comprehensive collection of annotated lung CT scans. The dataset was used for the LUNA16 challenge, which aimed to evaluate various algorithms for automated lung nodule detection. LUNA16 includes 888 low-dose thoracic CT scans, where each scan is accompanied by annotations from a panel of four experienced thoracic radiologists). The dataset's detailed annotations provide essential ground truth information for region of interest (ROI) segmentation, along with precise 3D coordinates of the nodules.

[1] https://luna16.grand-challenge.org
[2] https://wiki.cancerimagingarchive.net/display/Public/LIDC-IDRI

In the context of the LUNA16 challenge, a lung nodule is defined as a round or irregular opacity with a diameter of at least 3 mm, with nodules in the dataset ranging from small (3 mm) to large (>30 mm).

The pre-processing phase involved normalizing the image intensities and adjusting the Hounsfield Units of the CT scans to prepare the data for effective segmentation. For the first phase, we used original CTs with the ground truth of lung parenchyma for training. Then, we generated a 3D cube with nodules at the corresponding coordinates (Figure 1), to ensure that each slice of segmented ROI matches with the generated mask during the training process .

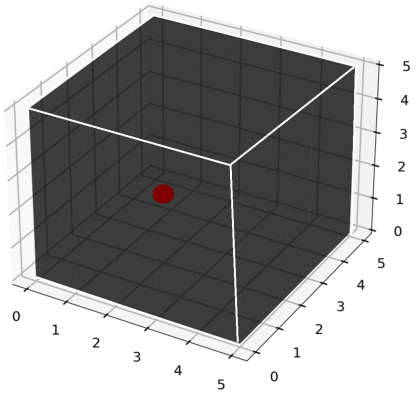


Fig. 1: Generated mask for Nodules.

2.1 LungQuant

For the first Phase (ROI) of the suggested algorithm, we have utilized Lung Quant v.2. LungQuant is a fully automated deep learning system developed to aid radiologists in identifying lung lesions associated with COVID-19 infection [2]. Initially launched in 2023, the system showed great potential, and a more refined version followed with improvements in segmentation accuracy for lung parenchyma and COVID-19 pneumonia in CT scans [3]. This section provides an in-depth look at the methodology behind LungQuant. Developed using a multi-step deep learning approach, LungQuant has been evaluated across various datasets to assess its effectiveness [6]. The process begins with an AlexNet-based DNN that predicts two key points to create a bounding box around the lungs in 3D voxel data, which aids in isolating the lung parenchyma. Following this, two U-net models are employed: the first segments the lung parenchyma (the approach applied in this project), while the second uses this segmentation to precisely identify and outline COVID-19 lesions. To enhance model performance and mitigate overfitting, pre-processing and data augmentation techniques were also incorporated. However, based on our specific needs, we utilized the first stage of this algorithm to segment the lung parenchyma as the ROI, removing unnecessary parts of the image to focus the model on detecting nodules more effectively. Results based on Dice Similarity Coefficeint demonstrate average of %90 for lung segmentation based on comparision with ground truth. The analysis of the samples indicates a high level of accuracy in the model's predictions in most cases. However, a few instances reveal erroneous segmentation or the omission of certain parts, resulting in lower dice scores.

3 Models

In this section, we will provide a detailed explanation of the models we tested, including the fine-tuned architectures specifically developed for our purpose.

3.1 Res U-Net

Res U-Net is a variant of the U-Net architecture that incorporates residual connections, a concept introduced in ResNet (Residual Network). Residual connections enable the model to focus on

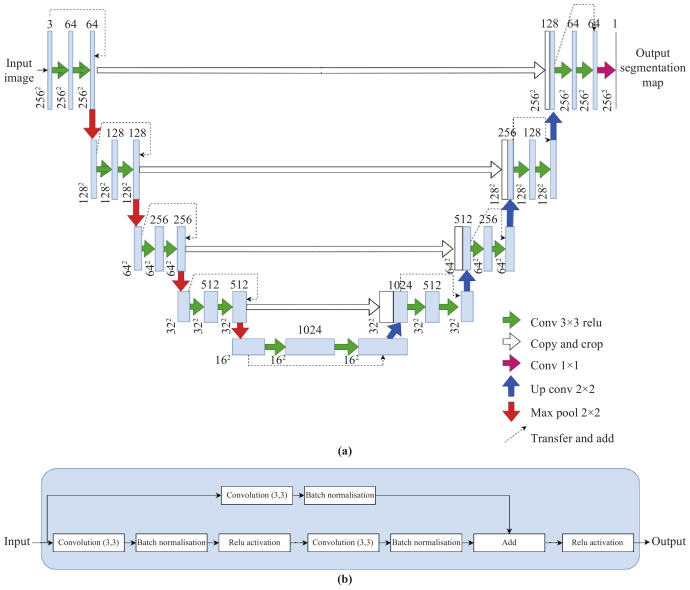


Fig. 2: Res U-net.

Layer Name	Output Shape	Description
Input	(256, 256, 1)	Input grayscale image
ConvBlock-32	(256, 256, 32)	Convolutional block with 32 filters
MaxPooling	(128, 128, 32)	Max pooling to reduce dimensions
ResConvBlock-64	(128, 128, 64)	Residual convolutional block with 64 filters
MaxPooling	(64, 64, 64)	Max pooling to reduce dimensions
ResConvBlock-128	(64, 64, 128)	Residual convolutional block with 128 filters
MaxPooling	(32, 32, 128)	Max pooling to reduce dimensions
ResConvBlock-256	(32, 32, 256)	Residual convolutional block with 256 filters
MaxPooling	(16, 16, 256)	Max pooling to reduce dimensions
ResConvBlock-512	(16, 16, 512)	Residual convolutional block with 512 filters (bottleneck)
UpSampling	(32, 32, 512)	Upsampling layer to restore spatial dimensions
Concatenate	(32, 32, 768)	Skip connection with ResConvBlock-256 output
ResConvBlock-256	(32, 32, 256)	Residual convolutional block with 256 filters
UpSampling	(64, 64, 256)	Upsampling layer to restore spatial dimensions
Concatenate	(64, 64, 384)	Skip connection with ResConvBlock-128 output
ResConvBlock-128	(64, 64, 128)	Residual convolutional block with 128 filters
UpSampling	(128, 128, 128)	Upsampling layer to restore spatial dimensions
Concatenate	(128, 128, 192)	Skip connection with ResConvBlock-64 output
ResConvBlock-64	(128, 128, 64)	Residual convolutional block with 64 filters
UpSampling	(256, 256, 64)	Upsampling layer to restore spatial dimensions
Concatenate	(256, 256, 96)	Skip connection with ConvBlock-32 output
ResConvBlock-32	(256, 256, 32)	Residual convolutional block with 32 filters
Conv2D	(256, 256, 1)	Final 1x1 convolution layer for segmentation
BatchNormalization	(256, 256, 1)	Normalization of final output
Output	(256, 256, 1)	Segmentation mask with sigmoid activation

Table 1: Residual U-Net Architecture

learning residuals—the differences between the input and output—rather than learning the entire transformation directly.

The Res U-Net architecture builds on the traditional U-Net framework by integrating residual blocks [7]. Each residual block includes shortcut connections that add the input of the block directly to its output after the convolutional layers. This approach helps preserve important features and spatial information as data passes through the network, making it particularly useful for complex segmentation tasks (see Figure 2).

In the context of lung tumor segmentation, Res U-Net is used to process and analyze annotated medical images of the lungs. The model is trained to identify and segment tumor regions from the surrounding lung tissue.

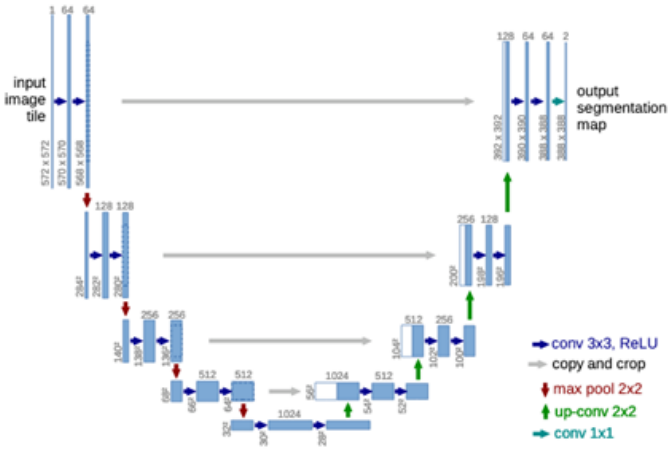


Fig. 3: Schematic of U-Net.

For our lung tumor segmentation project, Res U-Net is employed to handle the complexities of medical imaging. The model leverages its architectural features to accurately delineate tumors within lung images, supporting subsequent analysis and treatment planning.

The architecture of the Residual U-Net is outlined layer by layer in the table. The network starts with an input layer that accepts grayscale images of size $256 \times 256 \times 1$. In the downsampling path, a convolutional block with 32 filters is followed by residual convolutional blocks with progressively increasing filters (64, 128, 256, and 512), each accompanied by MaxPooling layers to reduce spatial dimensions by half at every stage. The bottleneck, located at the deepest layer of the network, uses a residual convolutional block with 512 filters to capture high-level features.

In the upsampling path, spatial dimensions are progressively restored using UpSampling layers, and feature maps are concatenated with their corresponding downsampling layers through skip connections. The upsampling path includes residual convolutional blocks with decreasing filter sizes (256, 128, 64, and 32), ensuring feature refinement at each stage. Finally, a 1×1 convolution layer with a sigmoid activation function produces the output segmentation mask, which matches the input size. This architecture incorporates residual connections to improve gradient flow and feature extraction, making it highly effective for image segmentation tasks.

3.2 Attention U-Net

Attention U-Net [4] is an advanced version of the traditional U-Net architecture, enhanced with attention mechanisms to improve its performance in image segmentation tasks. Attention, in the context of image segmentation, is a way to highlight only the relevant activations during training. This reduces the computational resources wasted on irrelevant activations, providing the network with better generalization power. Essentially, the network can pay "attention" to certain parts of the image, allowing it to focus more selectively on relevant parts while suppressing less important regions. This selective focus enables the model to better capture the essential features needed for accurate segmentation, particularly in medical imaging, where the boundaries of structures like tumors or organs can be subtle.

The attention mechanism dynamically adjusts to highlight areas of interest, leading to improved precision in segmenting complex structures. As a result, Attention U-Net is widely used in medical image segmentation tasks like lung nodule detection, tumor segmentation, and organ delineation. Its ability to focus on relevant image regions makes it particularly effective for tasks requiring high precision and sensitivity.

To understand why attention is beneficial in the U-Net, we need to look at the skip connections used in Figure 3. Grey arrows denote the long skip connections used in the U-Net [5]. During upsampling in the expanding path, spatial information recreated is imprecise. To counteract this problem, the U-Net uses skip connections that combine spatial information from the downsampling path with the upsampling path. However, this brings across many redundant low-level feature extractions, as feature representation is poor in the initial layers.

Attention Gateway. Attention gateways are mechanisms used in neural networks, particularly in attention-based models like Attention U-Net, to control and enhance the flow of information

through the network. These gateways allow the network to focus selectively on the most relevant parts of the input data, filtering out less important or irrelevant information. This selective focus improves the network's efficiency and accuracy, especially in tasks like image segmentation.

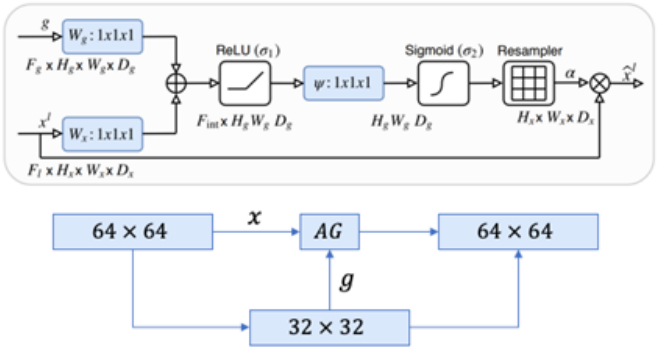


Fig. 4: Attention Gateway

Input Filtering. Attention gateways receive input features from the network and apply a filtering process to determine which parts of the input are most relevant to the task at hand. This is done by calculating an attention map, which assigns higher weights to the features that are deemed important and lower weights to those that are not.

Dynamic Focus. The attention map dynamically adjusts based on the context of the data, allowing the network to "pay attention" to different regions of the input depending on what is most important for the specific segmentation task. For example, in medical imaging, attention gateways might focus more on regions where abnormalities like tumors are likely to be present while ignoring background noise.

Enhancing Model Performance. By directing the network's focus to critical areas, attention gateways help reduce the computational load and prevent the network from being distracted by irrelevant information. This leads to better generalization, as the model becomes more adept at identifying and segmenting important structures within the image.

The table 2 outlines the architecture of the Attention U-Net, detailing its components layer by layer. The downsampling path comprises convolutional blocks (ConvBlock) with progressively increasing filter sizes (64, 128, 256, and 512) interspersed with MaxPooling layers to reduce the spatial dimensions by half at each stage. The bottleneck, located at the deepest part of the network, uses a convolutional block with 512 filters to extract high-level features. In the upsampling path, spatial dimensions are progressively restored using upsampling layers (UpBlock) combined with skip connections from the corresponding downsampling layers. Attention gates are applied at each skip connection to emphasize relevant features, improving segmentation accuracy. The network concludes with an output layer that applies a sigmoid activation function to produce a segmentation mask of the same size as the input. This architecture leverages attention mechanisms to refine feature fusion and convolutional blocks for effective feature extraction, making it well-suited for medical image segmentation tasks.

3.3 Attention Res U-Net

Attention Res-UNet [1] is an advanced deep-learning model designed for medical image segmentation, integrating residual networks (ResNet), the UNet architecture, and attention mechanisms. It extracts features through the encoder and reconstructs the segmented image in the decoder. By incorporating residual blocks from ResNet, the model addresses challenges like vanishing gradients and allows for deeper and more effective feature extraction. This helps the model learn more complex patterns, improving its ability to accurately detect small or subtle structures. The attention mechanism further enhances the model by selectively focusing on important regions of the input image, such as areas where lung nodules might appear while reducing the influence of irrelevant

Layer Name	Output Shape	Description
Input	(256, 256, 1)	Input grayscale image
ConvBlock-64	(256, 256, 64)	Convolutional block with 64 filters
MaxPooling	(128, 128, 64)	Max pooling to reduce dimensions
ConvBlock-128	(128, 128, 128)	Convolutional block with 128 filters
MaxPooling	(64, 64, 128)	Max pooling to reduce dimensions
ConvBlock-128	(64, 64, 128)	Convolutional block with 128 filters
MaxPooling	(32, 32, 128)	Max pooling to reduce dimensions
ConvBlock-256	(32, 32, 256)	Convolutional block with 256 filters
MaxPooling	(16, 16, 256)	Max pooling to reduce dimensions
ConvBlock-512	(16, 16, 512)	Convolutional block with 512 filters
AttentionGate-256	(16, 16, 256)	Attention gate for skip connection
UpBlock-256	(32, 32, 256)	Upsampled with 256 filters
AttentionGate-128	(32, 32, 128)	Attention gate for skip connection
UpBlock-128	(64, 64, 128)	Upsampled with 128 filters
AttentionGate-128	(64, 64, 128)	Attention gate for skip connection
UpBlock-128	(128, 128, 128)	Upsampled with 128 filters
AttentionGate-64	(128, 128, 64)	Attention gate for skip connection
UpBlock-64	(256, 256, 64)	Upsampled with 64 filters
Output	(256, 256, 1)	Output segmentation mask (sigmoid activation)

Table 2: Attention U-Net Architecture

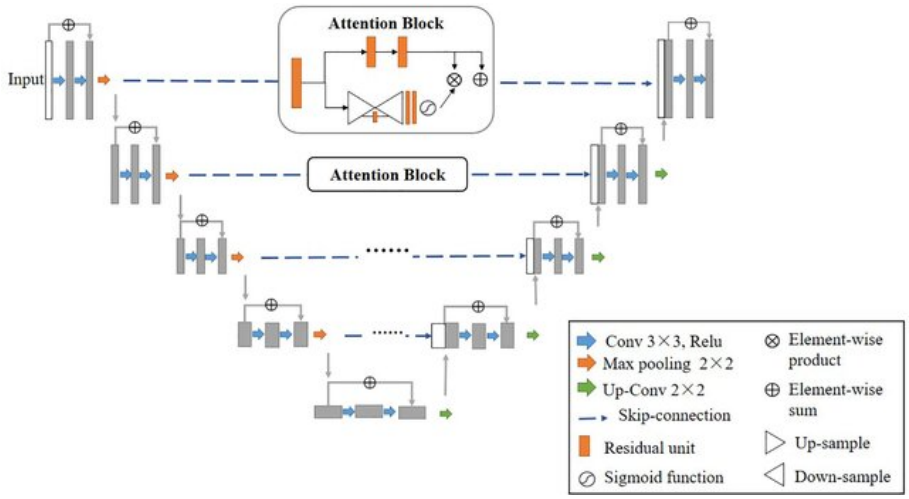


Fig. 5: Residual Attention U-Net

background regions. This makes the model more precise, especially in cases where the target structures are small or hard to distinguish. By combining these elements, Attention Res-UNet achieves high accuracy and robustness in medical image segmentation tasks, making it particularly effective for detecting lung nodules, where both precision and detail are crucial. The table 3 describes the architecture of the Attention Residual U-Net, highlighting each layer's role in the network. The downsampling path consists of residual convolutional blocks (ResBlocks) with progressively increasing filter sizes (32, 64, 128, 256, and 512) and MaxPooling layers that reduce the spatial dimensions by half after each ResBlock. At the bottleneck, the deepest ResBlock contains 512 filters. The upsampling path mirrors the downsampling path, starting with upsampling layers to restore spatial dimensions, followed by UpBlocks that concatenate the upsampled feature maps with corresponding skip connections from the downsampling path. Attention gates refine the skip connections by highlighting relevant regions, ensuring efficient feature fusion at each stage. The network concludes with an output layer that applies a sigmoid activation function to produce the final segmentation mask of the same size as the input, $256 \times 256 \times 1$. This architecture leverages residual connections for better gradient flow and attention mechanisms to improve segmentation accuracy by focusing on important features.

Layer Name	Output Shape	Description
Input	(256, 256, 1)	Input grayscale image
ResBlock-32	(256, 256, 32)	Residual block with 32 filters
MaxPooling	(128, 128, 32)	Max pooling to reduce dimensions
ResBlock-64	(128, 128, 64)	Residual block with 64 filters
MaxPooling	(64, 64, 64)	Max pooling to reduce dimensions
ResBlock-128	(64, 64, 128)	Residual block with 128 filters
MaxPooling	(32, 32, 128)	Max pooling to reduce dimensions
ResBlock-256	(32, 32, 256)	Residual block with 256 filters
MaxPooling	(16, 16, 256)	Max pooling to reduce dimensions
ResBlock-512	(16, 16, 512)	Residual block with 512 filters
AttentionGate-256	(16, 16, 256)	Attention gate for skip connection
UpBlock-256	(32, 32, 256)	Upsampled with 256 filters
AttentionGate-128	(32, 32, 128)	Attention gate for skip connection
UpBlock-128	(64, 64, 128)	Upsampled with 128 filters
AttentionGate-64	(64, 64, 64)	Attention gate for skip connection
UpBlock-64	(128, 128, 64)	Upsampled with 64 filters
AttentionGate-32	(128, 128, 32)	Attention gate for skip connection
UpBlock-32	(256, 256, 32)	Upsampled with 32 filters
Output	(256, 256, 1)	Output segmentation mask (sigmoid activation)

Table 3: Attention Residual U-Net Architecture

4 Metrics

Described models are run with the same dataset to compare the performance of the neural networks in the specific task. It is important to note that the initial phase of the proposed algorithm, which involves segmenting the region of interest, is handled by a pre-trained U-Net. Outputs of the first phase will not be a part of the comparison since it is consistent across all models. To effectively compare the results of these models, we will rely on four standard evaluation metrics.

DICE Similarity Coefficient. The Dice Similarity Coefficient (DSC) is a metric used to assess spatial overlap and validate the reproducibility of the results. The DSC value ranges between 0 (no overlap) and 1 (perfect overlap).

$$DSC = \frac{2TP}{2TP + FP + FN} \tag{1}$$

Sensitivity. Sensitivity, also known as recall or true positive rate (TPR), is a metric used in binary classification tasks, including medical image segmentation, to measure the model's ability to correctly identify positive instances. In the context of nodule detection or segmentation, sensitivity quantifies how effectively the model detects actual nodules among all nodules present in the dataset. It focuses on minimizing false negatives (FN) (missed nodules) rather than false positives (FP).

$$Sensitivity = \frac{TP}{TP + FN} \tag{2}$$

Precision. Precision is a metric that measures the accuracy of positive predictions by a model. It is calculated as the ratio of true positive (TP) predictions to the total number of positive predictions (true positives plus false positives). In other words, precision tells us how often the model is correct when it predicts a positive class. A high precision means that the model has a low rate of false positives, so when it predicts a positive result, it is likely correct.

Sensitivity and specificity together offer a more detailed understanding of a model's performance by differentiating between its ability to detect the target condition when it is present (sensitivity) and ratio of its ability to correctly recognize (Precision).

$$Precision = \frac{TP}{TP + FN} \tag{3}$$

False Positive Rate. The average false positive rate per scan (FPR) refers to the average number of times a model incorrectly identifies non-nodule regions as nodules in a single scan. This metric is important in medical imaging tasks, as it helps quantify the model's tendency to produce false alarms. In the context of lung nodule detection, it measures how often the model wrongly marks a normal area of the lung as a nodule in a given scan.

$$FalsePositiveRateperScan = \frac{FP}{numberofscans} \tag{4}$$

F1 Score. F1 Score evaluates the balance between precision and recall (sensitivity) in classification tasks. It is the harmonic mean of precision and recall, giving a single score that considers both false positives (FP) and false negatives (FN).

$$F1Score = 2.\frac{Precision.Recall}{Precision + Recall} \tag{5}$$

5 Experiments and Results

So far, we have discussed three different neural network models, as well as the LUNA-16 dataset used in our project. Our ultimate objective is to compare the performance of these models in the task of nodule segmentation using standard evaluation metrics. In this section, we will provide both visual and numerical results for each model, carefully examining their performance. Additionally, we will analyze the strengths and weaknesses of each approach, highlighting how certain models excel in specific areas of segmentation while others may fall short. This comparison will help us understand the trade-offs and benefits of each model in the context of lung nodule detection.

Models	Dice	Sensitivity	Precision	FPR	F1 Score
Res U-Net	83	73	78	2.44	75
Attention U-Net	86	85	88	3.16	86
Attention Res U-net	90	88	92	0.5	88

Table 4: Numerical Results of tested models

The evaluation results in Table 4 demonstrate that the Attention Res U-Net outperforms the other models across most metrics, making it the most reliable choice for this segmentation task. With the highest Dice score (90), it achieves the best overall segmentation quality, reflecting superior overlap between predictions and ground truth. It also excels in precision (92) and F1 score (88), indicating its ability to accurately detect true positives while minimizing false positives, which is critical for reducing unnecessary follow-ups in clinical applications. Moreover, its False Positive Rate (FPR) of 0.5 is significantly lower than both the Res U-Net and Attention U-Net, further emphasizing its specificity. While the Attention U-Net achieves slightly higher sensitivity (85) compared to Attention Res U-Net (88), its higher FPR (3.16) suggests a trade-off in specificity. The Res U-Net, with a Dice score of 83, lags behind both attention-based models, highlighting the impact of incorporating attention mechanisms and residual connections for improved performance. Overall, the Attention Res U-Net strikes the best balance between sensitivity, precision, and specificity, making it the most effective and reliable model for this task.

In addition, Figure 6 compares the performance of three deep learning models—Res U-Net, Attention U-Net, and Attention res U-Net—on lung nodule segmentation using a test set of CT scans. Each row represents a different test case, starting with the original CT image, followed by the segmented ROI, and the ground truth mask that indicates the true locations of nodules. The next three columns display the predictions made by the models. Res U-Net, incorporating residual connections, shows varying results, with some instances missing nodules or capturing only a small portion. Attention U-Net, which uses an attention mechanism to focus on relevant parts of the image, generally shows better localization of nodules, though occasional misses or less defined detections are still observed.

The Attention res U-Net model consistently delivers the most accurate results across the test cases. This model's combination of residual connections and attention mechanisms enhances its

ability to detect subtle features, allowing it to identify nodules more reliably. By preserving essential features through residual connections and refining the focus through attention, this model produces more accurate segmentation, particularly in challenging cases. Overall, the results indicate that Attention res U-Net offers superior performance compared to the other models, striking a better balance between sensitivity and precision in lung nodule segmentation.

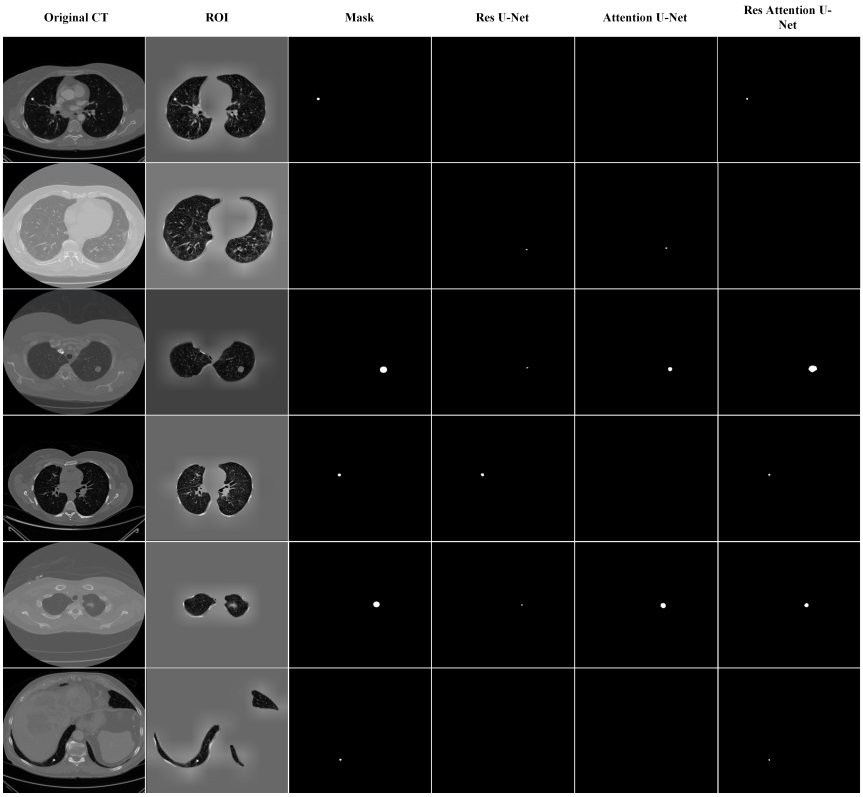


Fig. 6: Samples of test set with tested models

Moreover, Figure 7 continues this comparison across additional test cases, further reinforcing the trends observed in Figure 6. Attention res U-Net once again demonstrates the most consistent and precise nodule detection, while the other models tend to either miss smaller nodules or detect them less clearly. This figure supports the conclusion that integrating both attention mechanisms and residual connections enhances the model's capability, making Attention res U-Net the most effective among the three for accurate nodule segmentation.

6 Conclusion

In this study, we evaluated three deep learning models—Res U-Net, Attention U-Net, and Attention Res U-Net—for the challenging task of lung nodule segmentation on the LUNA-16 dataset. Through our experiments, we demonstrated that the inclusion of attention mechanisms and residual connections significantly enhances segmentation performance. Among the tested models, the Attention Res U-Net achieved the best results, with a Dice score of 90, precision of 92, sensitivity of 88, and the lowest false positive rate of 0.5, indicating its robustness in accurately segmenting lung nodules while minimizing false detections.

The comparative analysis revealed that while the Res U-Net provided a strong baseline, it struggled with nodule detection in complex cases due to its reliance on simpler architectural features. The Attention U-Net introduced attention mechanisms that improved sensitivity and precision, though at the cost of a slightly higher false positive rate. The Attention Res U-Net, by combining attention mechanisms with residual connections, achieved a superior balance between sensitivity, precision, and specificity, making it the most effective model for this task.

These findings highlight the importance of designing architectures tailored to the specific demands of medical image segmentation. Furthermore, integrating advanced mechanisms like atten-

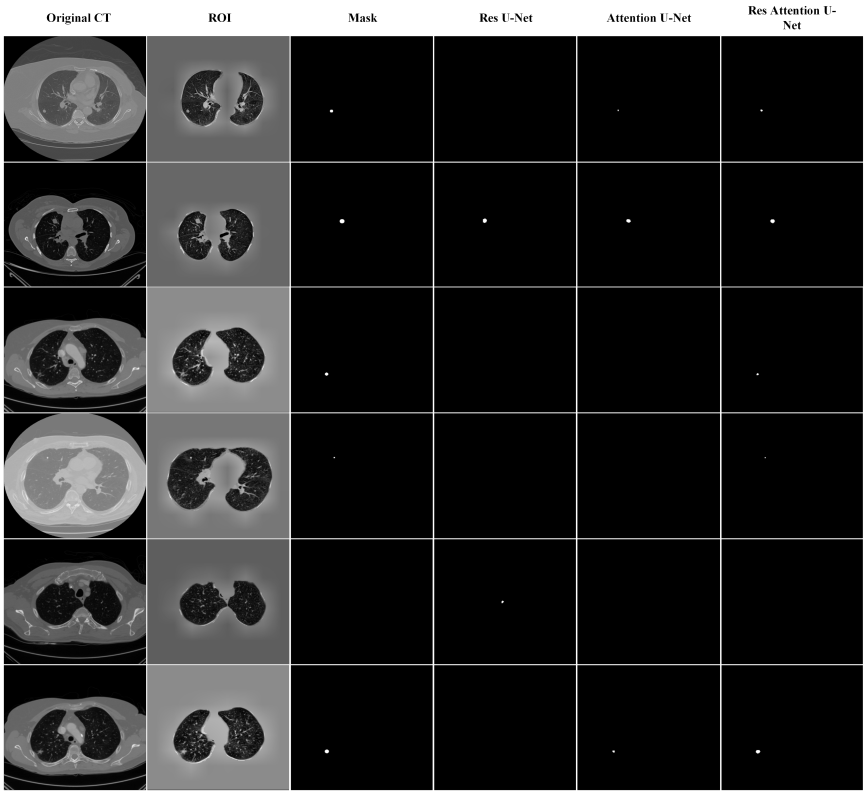


Fig. 7: Samples of test set with tested models

tion gates and residual connections provides significant advantages in detecting small or subtle features, such as lung nodules, which are critical for early diagnosis and treatment planning.

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